LF5 Worked Examples: Linear Models

Recording companion for the LF5 CheckIt support video — Math 172, Santa Ana College

How to use this document

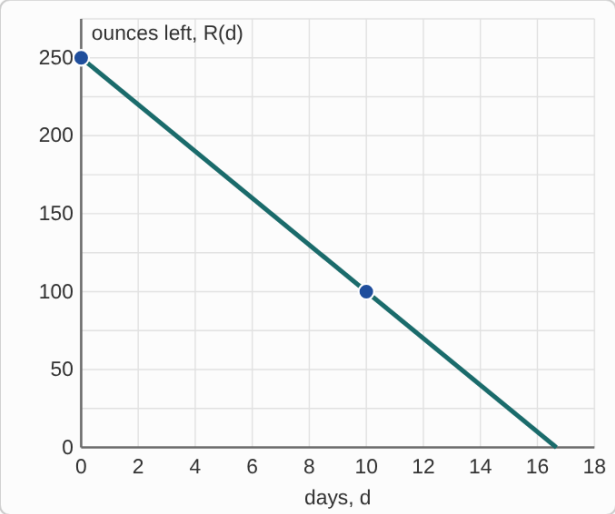
- Each worked example is on its own page, in the same order as the Try It tabs in the interactive explainer, with the same numbers.
- The right-hand third of every page is intentionally blank. It is space for handwriting or ink notation while you narrate (for example, after inserting this PDF as a printout in OneNote).
- Say it at the start of the video: your CheckIt problem will have different numbers than the examples here. The reasoning is what transfers.

The one question behind every example

What is the starting amount, and what changes by the same amount each time? A linear model is a starting amount plus a constant rate times the input: $f(x) = mx + b$. The rate is the slope, with a minus sign if the amount goes down. The starting amount is the y-intercept, the output when the input is 0. Before you plug in or solve, check what x counts: items, additional lines, days, or years after the start year.

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Example 1: A Bag of Rice



Plotted data: A graph with days, d, from 0 to 18 across and ounces left, R(d), from 0 to 275 up. A straight line starts at (0, 250), marked with a dot, falls through the marked point (10, 100), and reaches 0 between day 16 and day 17.

Question

A family buys a 250-ounce bag of rice and uses 15 ounces per day. R(d) is the rice left after d days. What is the slope of R, and what does it mean?

A. 250 ounces: the amount in the bag at the start
B. −15 ounces per day: the rice left drops 15 ounces a day
C. −15 ounces: the amount of rice that is left after one day
D. 15 days: how long the bag of rice lasts before it is empty

Answer and reasoning

The model is $R(d)=250-15d$. Each day the rice left goes down by 15 ounces, so the slope is -15, in ounces per day. On the graph, the line drops 150 ounces over 10 days: $(-150)/10=-15$. CheckIt's answer key prints this slope as 15, the rate of use; say both: the family uses 15 ounces a day, so the slope of R is -15. The 250 ounces is the y-intercept, not the slope; the rice left after one day is $R(1)=235$ ounces, an amount,

while the slope is a rate; and the bag lasts $250/15 \approx 16.7$ days, not 15, because 15 is the ounces used per day. **−15 ounces per day.**

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Example 2: A Phone Plan

$$C(x)=10x+35$$

Question

A phone plan costs $35 a month for the first line and $10 for each additional line. C(x) is the monthly cost with x additional lines. What is the y-intercept, and what does it mean?

A. 35: the cost with only the first line
B. 35: the cost of each additional line on the plan
C. 45: the cost of the first line, since x = 1
D. 0: a plan with no lines costs nothing

Answer and reasoning

The model is $C(x)=10x+35$, and the y-intercept is $C(0)=35$. Here x counts additional lines, so $x=0$ is a plan with just the first line, which costs $35. Each additional line is $10, the slope, not 35; $C(1)=45$ is the plan with one additional line, two lines in all, because $x=1$ does not mean the first line; and $x=0$ still includes the first line, so the cost is not 0. **35: the cost with only the first line.**

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Example 3: A Shrinking Town

$$P(x)=48000-1500x$$

Question

In 2010, the population of Pawnee was 48,000, and it has decreased by 1,500 per year since. P(x) is the population x years after 2010. What is the y-intercept, and what does it mean?

A. 48,000: the population in the year 0
B. 2010: the year the model starts
C. −1,500: the change in population each year
D. 48,000: the population in 2010

Answer and reasoning

The model is $P(x)=48{,}000-1{,}500x$, and the y-intercept is $P(0)=48{,}000$. Since x counts years after 2010, $x=0$ is the year 2010 itself. So 48,000 is the population in 2010, not in the calendar year 0; 2010 is the start year, which tells you what $x=0$ means but is not an output of the model; and $-1{,}500$ people per year is the slope. **48,000: the population in 2010.**

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Example 4: Printing Posters

$$C(x)=3x+150$$

Question

Producing posters costs $3 per poster plus an additional $150 for fixed costs. C(x) is the cost of producing x posters. What is the slope, and what does it mean?

A. 3: each extra poster adds $3
B. 150: the cost goes up $150 for each poster
C. 153: the cost of producing the first poster
D. 1/3: each extra dollar makes 1/3 of a poster

Answer and reasoning

The model is $C(x)=3x+150$. The slope is the rate, 3 dollars per poster: each additional poster adds $3. The $150 is paid once, whatever the number of posters, so it is the *y*-intercept, not a rate; 153 is $C(1)$, the total cost of one poster, which includes the fixed costs; and $1/3$ poster per dollar flips the rate, input over output. **3 dollars per poster.**

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Example 5: Widgets

$$C(x)=mx+b$$

Question

Manufacturing widgets costs $12 per widget plus an additional $400 for fixed costs. Which function gives C(x), the cost of manufacturing x widgets?

A. $C(x)=400x+12$
B. $C(x)=12x+400$
C. $C(x)=12x-400$
D. $C(x)=412x$

Answer and reasoning

The rate is $12 per widget, so the slope is 12; the fixed costs, $400, are paid even for 0 widgets, so they are the y-intercept. $C(x)=12x+400$. Check: 0 widgets cost $C(0)=400$, the fixed costs. $400x+12$ swaps the rate and the starting amount; $12x-400$ subtracts the fixed costs, but they add to the cost; and $412x$ adds the two numbers, which charges the fixed costs again for every widget. ***C(x)=12x+400*.**

Example 6: Population

$$P(x)=mx+b$$

Question

In 2015, the population of Springfield was 85,000, and it has decreased by 3,000 per year since. Which function gives P(x), the population x years after 2015?

A. $P(x)=3000x+85000$
B. $P(x)=85000-3000x$
C. $P(x)=85000x-3000$
D. $P(x)=85000-3000(x-2015)$

Answer and reasoning

The town starts at 85,000 when $x=0$, the year 2015, and loses 3,000 people each year, so the slope is $-3{,}000$: $P(x)=85{,}000-3{,}000x$. $3000x+85000$ makes the population grow; $85000x-3000$ swaps the rate and the starting amount; and $85000-3000(x-2015)$ treats x as the calendar year, but here x is years after 2015, so it would give $P(0)=6{,}130{,}000$. ***P(x)=85{,}000−3{,}000x*.**

Example 7: Phone Lines

$$C(x)=mx+b$$

Question

A phone plan costs $25 a month for the first line and $15 for each additional line. Which function gives C(x), the monthly cost with x additional lines?

- **A.** $C(x)=15x+25$
- **B.** $C(x)=25x+15$
- **C.** $C(x)=15(x-1)+25$
- **D.** $C(x)=40x$

Answer and reasoning

The first line is always there, so $25 is the starting amount, and each additional line adds $15: $C(x)=15x+25$. Check: one additional line, two lines in all, costs $C(1)=40$. $25x+15$ swaps the two prices; $15(x-1)+25$ would be right if x counted all the lines, but x counts only the additional ones; and $40x$ charges both prices for every additional line. ***C(x)=15x+25*.**

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Example 8: A Leaking Tank

$$W(x)=mx+b$$

Question

A leaky tank is losing water at a constant rate of 4 ounces per hour. The tank started with 200 ounces. Which function gives W(x), the water left after x hours?

A. $W(x)=4x+200$
B. $W(x)=200x-4$
C. $W(x)=200-4x$
D. $W(x)=4x-200$

Answer and reasoning

The tank starts with 200 ounces and loses 4 each hour, so the slope is -4: $W(x)=200-4x$. Check: after 10 hours, $200-40=160$ ounces are left. $4x+200$ adds water instead of losing it; $200x-4$ swaps the rate and the starting amount; and $4x-200$ subtracts in the wrong order, which starts the tank at -200 ounces. ***W(x)=200−4x*.**

Example 9: A Calendar Year

$$P(x)=34000+2000x$$

Question

In 2012, the population of Capeside was 34,000, and it has grown by 2,000 per year since. P(x) is the population x years after 2012. What will the population be in 2027?

A. 4,088,000
B. 64,000
C. 30,000
D. 62,000

Answer and reasoning

The model is $P(x)=34{,}000+2{,}000x$. The year 2027 is $x=2027-2012=15$ years after the start, so $P(15)=34{,}000+30{,}000=64{,}000$. Putting in $x=2027$ gives 4,088,000, because x is years after 2012, not the year; 30,000 is only the growth, $2{,}000(15)$, without the starting population; and 62,000 uses 14 years, counting from 2013 instead of 2012. **64,000.**

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Example 10: How Many Hats

$$C(x)=9x+480$$

Question

Producing hats costs $9 per hat plus an additional $480 for fixed costs. How many hats can be produced for $1,830?

A. 150 hats
B. about 203 hats
C. about 257 hats
D. 16,950 hats

Answer and reasoning

The model is $C(x)=9x+480$, and this time the output is known: solve $9x+480=1830$. Subtract 480: $9x=1350$. Divide by 9: $x=150$. Check: $9(150)+480=1830$. About 203 is $1830/9$, which forgets the fixed costs; about 257 is $(1830+480)/9$, which adds 480 instead of subtracting it; and 16,950 is $C(1830)$, which puts the dollars in as the number of hats. **150 hats.**

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Example 11: Rice Left

$$R(d)=300-12d$$

Question

A family buys a 300-ounce bag of rice and uses 12 ounces per day. After how many days will the family have 84 ounces of rice left in the bag?

A. 7 days
B. 32 days
C. −708 days
D. 18 days

Answer and reasoning

The model is $R(d)=300-12d$. Solve $300-12d=84$: subtract 300, $-12d=-216$, and divide by -12, $d=18$. Check: $300-12(18)=84$. 7 days is $84/12$, which divides the amount left instead of the amount used; 32 days is $(300+84)/12$, which adds the 84; and -708 is $R(84)$, which puts the ounces in as the number of days. **18 days.**

Example 12: Plan Cost

$$C(x)=5x+30$$

Question

A phone plan costs $30 a month for the first line and $5 for each additional line. How much does a plan with 4 additional lines cost?

A. $45
B. $50
C. $140
D. $120

Answer and reasoning

The model is $C(x)=5x+30$, with x the number of additional lines. $C(4)=5(4)+30=50$. $45 uses $x=3$, counting the first line as one of the 4; $140 is $4(30+5)$, which charges both prices for every line; and $120 is $4(30)$, which charges the first-line price for each additional line. **$50.**

Example 13: Slip 1

$$W(x)=3x+55$$

Question

A leaky bucket starts with 55 ounces of water and loses 3 ounces per minute. A student wrote this model for W(x), the water left after x minutes. What is the mistake, if any?

A. There is no mistake.
B. Water is lost, so the slope is −3: W(x) = 55 − 3x.
C. The rate and the start are swapped: it should be W(x) = 55x + 3.
D. Subtract the 55 instead: W(x) = 3x − 55, since water is lost.

Answer and reasoning

The bucket loses water, so the amount goes down 3 ounces each minute and the slope is -3: $W(x)=55-3x$. The student's model has the bucket gaining water. $55x+3$ swaps the rate and the starting amount; and $3x-55$ subtracts in the wrong order, which starts the bucket at -55 ounces. ***$W(x)=55-3x$*.**

Example 14: Slip 2

$$P(x)=72000+4000x$$
$$P(2023)=72000+4000(2023)$$
$$=8164000$$

Question

In 2018, the population of Bayside was 72,000, and it has grown by 4,000 per year since. P(x) is the population x years after 2018. A student found the population in 2023. What is the mistake, if any?

A. There is no mistake.

B. x counts years after 2018: use x = 5, so P(5) = 92,000.

C. The town grows 4,000 once, so it should be 76,000.

D. Count 2018 through 2023: use x = 6, so P(6) = 96,000.

Answer and reasoning

The model is right, but 2023 is not the input. The input is years after 2018: $x=2023-2018=5$, and $P(5)=72{,}000+20{,}000=92{,}000$. The student's 8,164,000 is far too big for a town that grows 4,000 a year. 76,000 is one year of growth, $P(1)$; and counting both 2018 and 2023 gives 6, but from 2018 to 2023 is 5 years, so 96,000 is the population in 2024. **92,000.**

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Example 15: Slip 3

$$C(x)=10x+40$$
$$C(3)=10(3)+40=70$$

Question

A phone plan costs \$40 a month for the first line and \$10 for each additional line. A student found the monthly cost with 3 additional lines. What is the mistake, if any?

A. There is no mistake.
B. The first line is not extra: use C(x) = 10(x − 1) + 40.
C. Three additional lines means x = 4, so the cost is \$80.
D. The slope should be 40, the bigger of the two prices.

Answer and reasoning

The first line costs \$40 whatever happens, so it is the starting amount, and each additional line adds \$10: $C(x)=10x+40$. With 3 additional lines, $C(3)=70$, which is 4 lines in all. The student is right. $10(x-1)+40$ would be right if x counted every line, but here x counts only the additional ones; for the same reason, 3 additional lines is $x=3$, not 4; and the slope is the price that repeats, \$10, not the larger price. **There is no mistake.**

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Example 16: Slip 4

$$C(x)=7x+600$$

Question

Manufacturing books costs $7 per book plus an additional $600 for fixed costs. A student wrote this model and said, "The y-intercept, 600, is the cost of manufacturing one book." What is the mistake, if any?

A. There is no mistake.
B. The y-intercept is 7, the cost of each book.
C. 600 is the cost of 0 books: the fixed costs.
D. 600 is the number of books that can be made, not a cost.

Answer and reasoning

The model is right, but the explanation is not. The *y*-intercept is the output when the input is 0: $C(0)=600$, the fixed costs paid before any books are made. One book costs $C(1)=607$. The 7 is the slope, the cost per book, not the *y*-intercept; and $C(x)$ is a cost in dollars, so 600 is dollars, not a number of books. **600 dollars is the cost of making 0 books: the fixed costs.**

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Example 17: Slip 5

$$180-9d=54$$
$$-9d=234$$
$$d=-26$$

Question

A family buys a 180-ounce bag of rice and uses 9 ounces per day. A student found when 54 ounces will be left. What is the mistake, if any?

A. There is no mistake.
B. Set the model equal to 0 instead, which gives d = 20 days.
C. Just divide the 54 by the 9 ounces a day: d = 6 days.
D. Subtracting 180 gives −9d = −126, so d = 14.

Answer and reasoning

The model $R(d)=180-9d$ and the equation are right. Then subtract 180 from both sides: $54-180=-126$, so $-9d=-126$ and $d=14$. The student added 180 instead, $54+180=234$. A negative number of days is a sign that something went wrong. Check: $180-9(14)=54$. Setting the model equal to 0 answers a different question, when the bag is empty; and $54/9=6$ divides the amount left instead of the amount used. **14 days.**